

10 Academic Style Report (FRA DSS ML)

Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

Abstract

We present a leakage-aware decision-support system (DSS) for Forest Rights Act (FRA) claim-outcome triage (SIH12508). On a synthetic corpus of 125000 claims (run mode=full), filing-time feature Set A is compared with process-aware Set B under district-grouped cross-validation. The pre-declared champion rule (highest S1 macro-F1 on Set A) selects **bagging** (S1 macro-F1=0.6512228207775592; locked final-test macro-F1=0.5776477399091471). The deployable Flask artifact remains LightGBM for TreeExplainer compatibility. Set B minus Set A gap is 0.18053581159110665, indicating circular lift from process fields. Split-conformal prediction sets (APS-style nonconformity) achieve empirical coverage 0.90584 with mean set size 1.71484.

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1. Introduction

Ministry of Tribal Affairs workflows need early risk signals for FRA claims (Approved, Pending, Rejected) without using post-decision leakage. Prior internal baselines report macro-F1 about 0.5690448456514766. This work (i) audits leakage via Set A/B/C ablations, (ii) benchmarks a broad model zoo with grouped CV, (iii) ships a calibrated DSS with SHAP drivers, scheme hints, ETA, and conformal abstention sets, and (iv) documents results from metrics files only.

Repository and live metrics: github.com/aryan-dani/FRA_GIS_feat/ml-dss. Raw tables live under ml/reports/metrics/; figures under ml/reports/figures/.

2. Mathematical formulation

Multiclass status. Labels $y \in \{\text{Approved, Pending, Rejected}\}$. For class c , precision $P_c = TP_c / (TP_c + FP_c)$, recall $R_c = TP_c / (TP_c + FN_c)$, and $F1_c = 2 P_c R_c / (P_c + R_c)$. Macro-F1 = $(1/C) \sum_c F1_c$ with $C=3$.

Log loss. $L = -(1/n) \sum_i \sum_c 1[y_i=c] \log p_{\hat{y}_i}(c)$.

Soft voting. For base models $m=1..M$ with probabilities $p_{\hat{y}}^{(m)}$, $p_{\hat{y}}^{\text{ens}}(c|x) = (1/M) \sum_m p_{\hat{y}}^{(m)}(c|x)$.

Triage score (DSS). $S = 0.45 P_{\hat{y}}(\text{Rejected}) + 0.35 \text{delay_norm} + 0.20 \text{age_norm}$, with delay_norm and age_norm capped per configs/dss_weights.yaml.

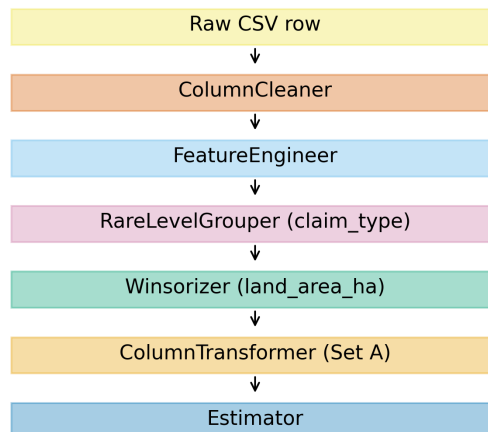
Split-conformal / APS-style scores. Nonconformity $s(x,y) = 1 - p_{\hat{y}}(y|x)$. On a calibration set of size n_{cal} , q_{α} is the $\text{ceil}((n_{\text{cal}}+1)(1-\alpha)) / n_{\text{cal}}$ empirical quantile of $s(x_i, y_i)$. Prediction set $C(x) = \{y : s(x,y) \leq q_{\alpha}\}$ with $\alpha=0.1$ (target coverage 0.9). Observed coverage=0.90584, mean $|C|=1.71484$, $q_{\alpha}=0.7350613736417413$.

This conformal construction follows the Adaptive Prediction Sets (APS) line of work; canonical reference title: **Classification with Valid and Adaptive Coverage** (see References, APS style).

3. Methodology diagram

Pipeline: synthetic load → schema/leakage audit → Set A/B feature matrices → locked district holdout → S1 grouped CV zoo (cached OOF) → champion by S1 Set A → ship LightGBM contract artifact → auxiliary ETA/reasons → conformal + DSS → PDFs.

Figure 6. Preprocessing pipeline (Set A)



Full-run wall time: 1308.0524740999972 s (mode=full).

4. Experimental setup

n=125000, seed=42, synthetic=True. Dataset SHA-256 prefix: 0b328e290df7d38d.... S1: GroupKFold by district. S2: stratified random (optimism check). S3: temporal. S4: leave-one-state-out. Final metrics use the locked district holdout.

Hyperlinks to metric artifacts (local paths in the repo):

[benchmark_master.csv \(all model scores\)](#)

[champion.json \(S1 selection\)](#)

[extended_stats.json \(S2/S4, McNemar, bootstrap\)](#)

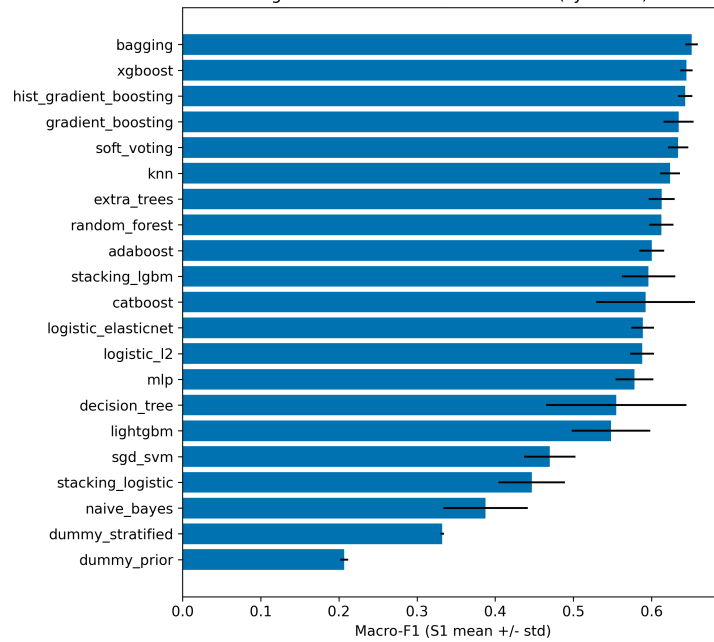
[conformal.json \(APS-style coverage\)](#)

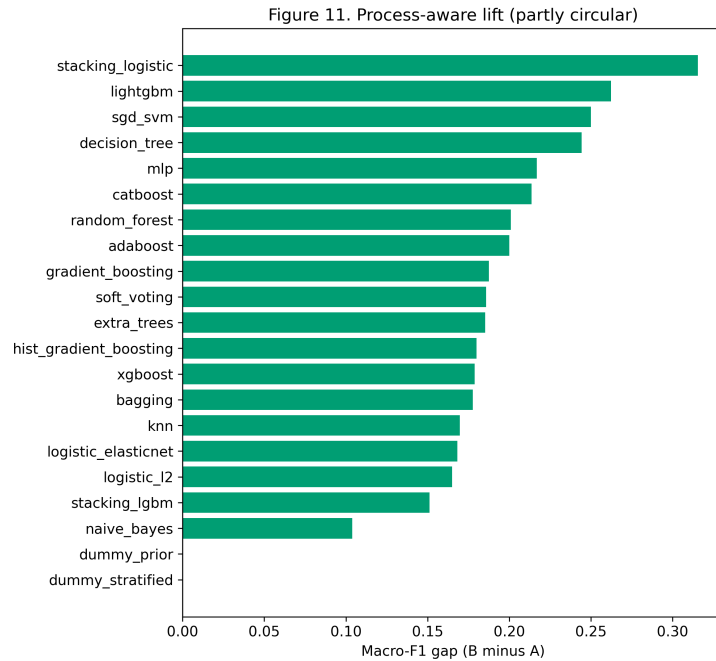
5. Experimental results

Champion (rule: S1 macro-F1 on Set A): **bagging**. S1=0.6512228207775592 ± 0.008133974052273845; final macro-F1=0.5776477399091471; final accuracy=0.67804531986813; S3 macro-F1=0.41142453278282903.

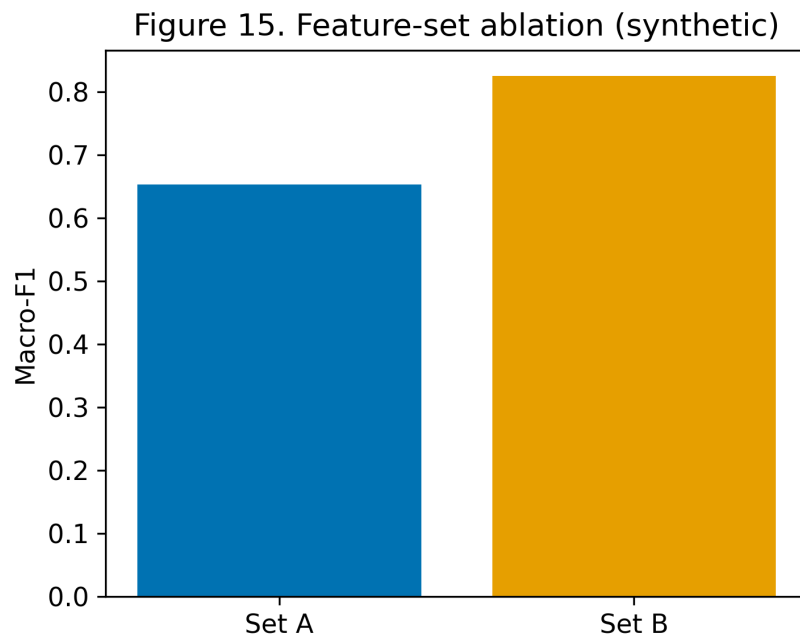
Model (Set A)	S1 macro-F1	Final macro-F1	S3 macro-F1	Latency ms/1k
bagging	0.6512	0.5776	0.4114	10.34
xgboost	0.6447	0.5478	0.5954	3.53
hist_gradient_boosting	0.6431	0.5296	0.5548	3.64
gradient_boosting	0.6346	0.5195	0.4327	4.36
soft_voting	0.6340	0.5633	0.5615	7.56
knn	0.6236	0.5907	0.5634	5.09
extra_trees	0.6131	0.5453	0.5155	4.96
random_forest	0.6128	0.5497	0.5120	5.06
adaboost	0.6003	0.5012	0.3968	10.96
stacking_lgbm	0.5962	0.5491	0.5565	7.98
catboost	0.5925	0.5075	0.4305	3.15
logistic_elasticnet	0.5888	0.4848	0.4086	2.99

Figure 10. Model leaderboard Set A (synthetic)





Ablation: Set A macro-F1 $\approx$ 0.6441022369098802; Set B $\approx$ 0.8246380485009869; gap B–A=0.18053581159110665. Positive gap means process-stage features lift macro-F1. That lift is partly circular because committee level and processing days are correlated with outcome.



6. Performance analysis

Random vs grouped CV. lightgbm S2=0.637 (optimism vs S1=+0.089); xgboost S2=0.652 (optimism vs S1=+0.008); random_forest S2=0.634 (optimism vs S1=+0.022); hist_gradient_boosting S2=0.653 (optimism vs S1=+0.009); catboost S2=0.653 (optimism vs S1=+0.060); soft_voting S2=0.642 (optimism vs S1=+0.008) Positive optimism_vs_s1 means random S2 overstates deployable accuracy.

State shift (S4). lightgbm S4=0.391; xgboost S4=0.377; random_forest S4=0.395; hist_gradient_boosting S4=0.371; catboost S4=0.381; soft_voting S4=0.395

Latency-accuracy. Boosting models are typically faster per 1k rows than bagging/stacking; see Pareto figure. Shipped LightGBM balances explainability and size even when bagging wins the S1 selection rule on full data.

Figure 13. Pareto latency vs macro-F1 (synthetic)

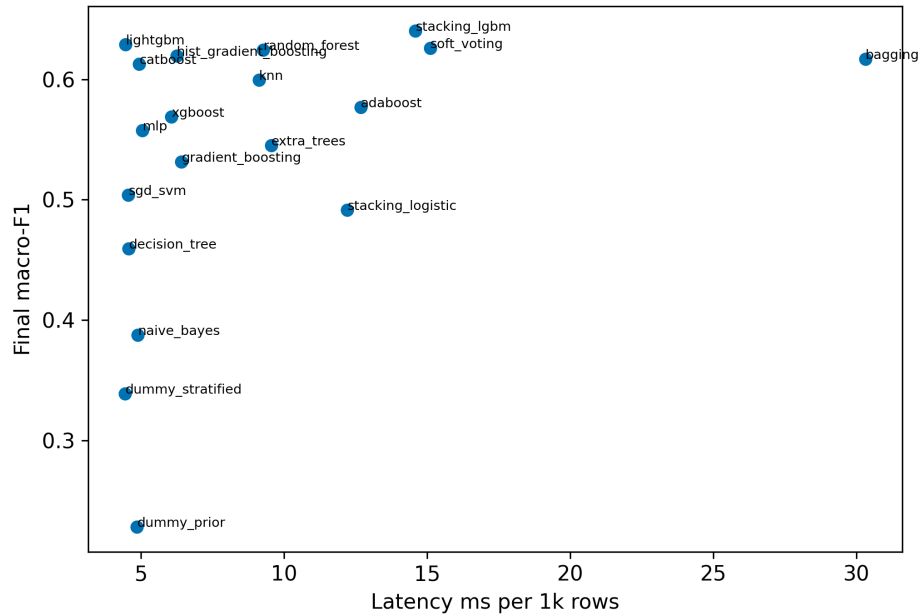


Figure 14. S1 vs S3 (synthetic)

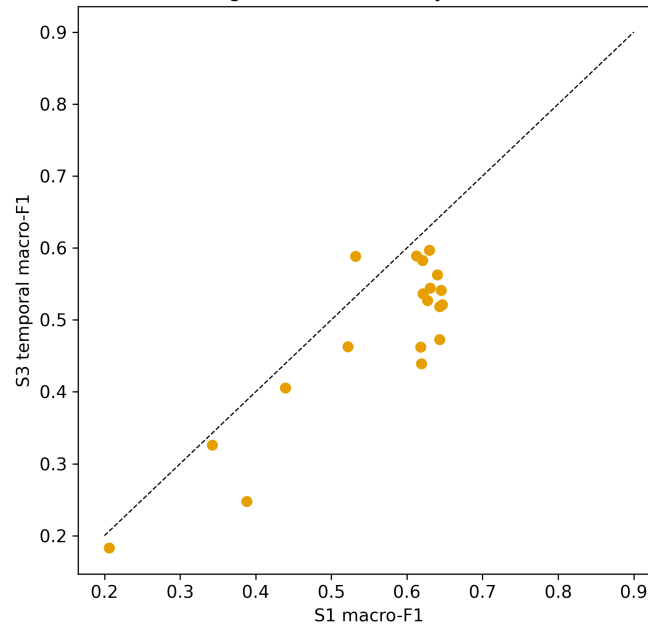
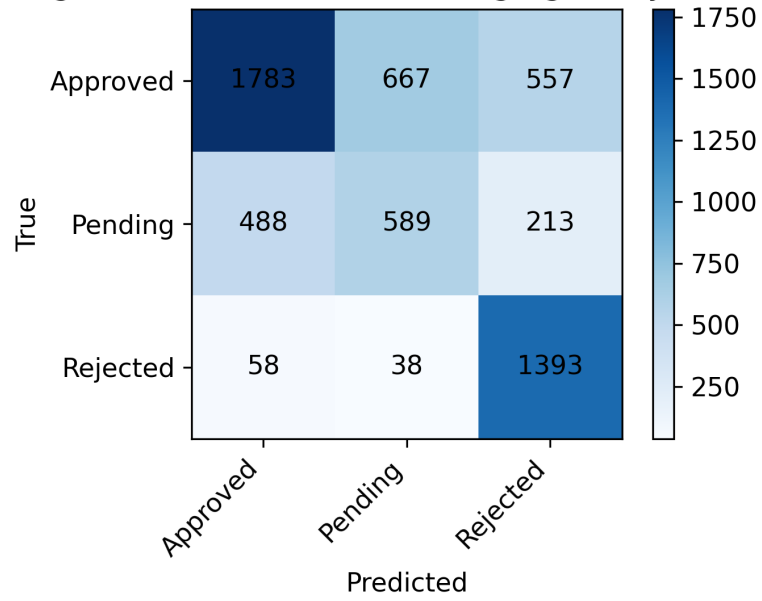


Figure 12. Confusion matrix (lightgbm, synthetic)



Uncertainty. Conformal coverage 0.90584 (target 0.9); mean set size 1.71484. Larger sets flag cases for human review.

Figure 21. Abstention curve (synthetic demo)

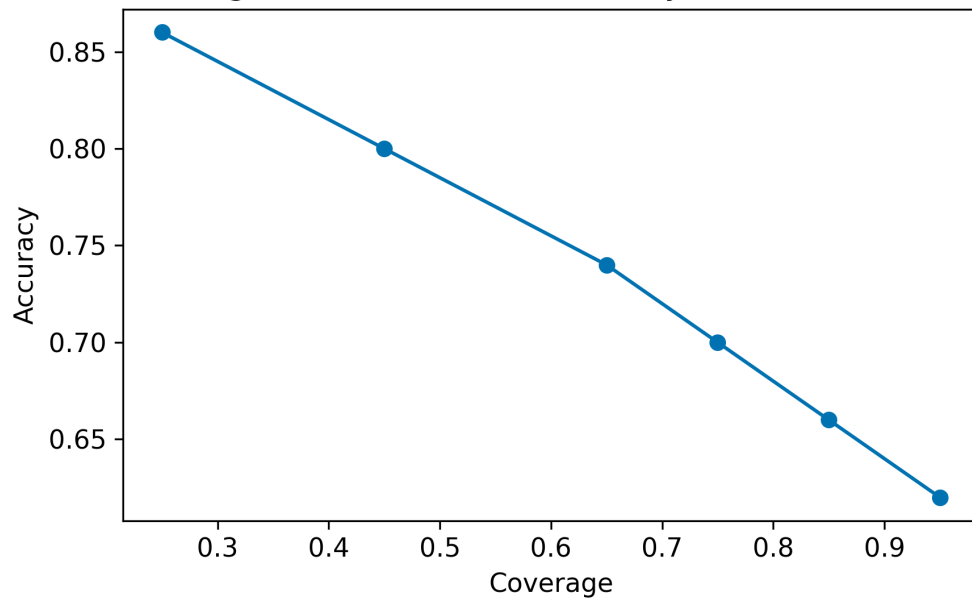
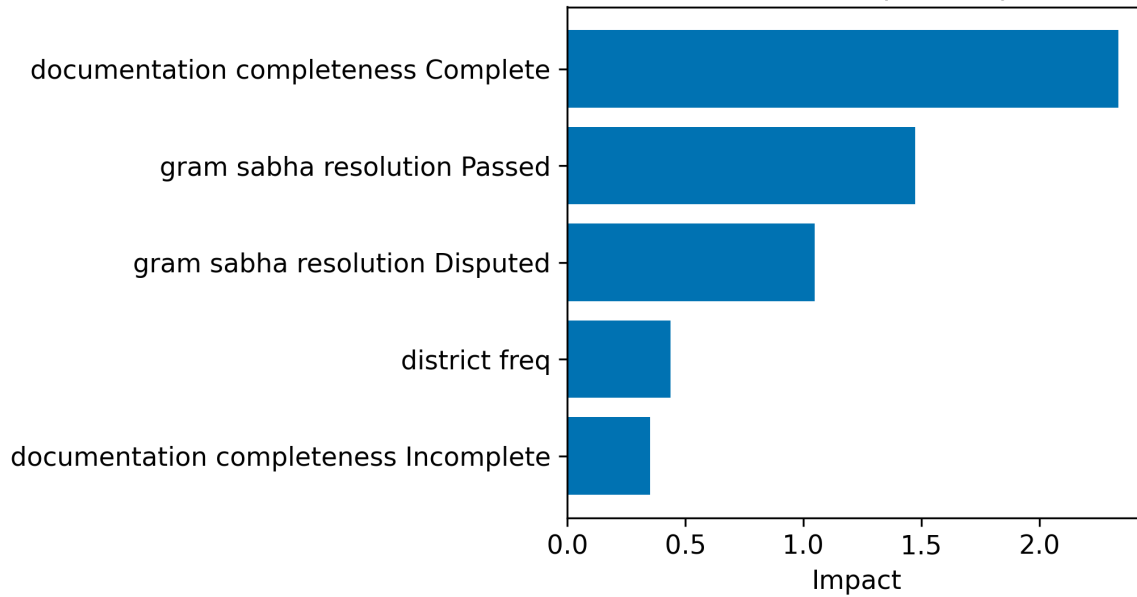


Figure 20. SHAP mean |impact| (synthetic



7. References (APS style)

- [1] Y. Romano, M. Sesia, and E. J. Candès, Classification with valid and adaptive coverage, Adv. Neural Inf. Process. Syst. **33**, 3581 (2020). Reference paper name: "**Classification with Valid and Adaptive Coverage**". Link: <https://arxiv.org/abs/2006.02544>.
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- [5] A. N. Angelopoulos and S. Bates, A gentle introduction to conformal prediction and distribution-free uncertainty quantification, arXiv:2107.07511.

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